

LinfoSec VMDR Practice Lab

Reference — Deploying Qualys Cloud Agent to EC2 via SSM and S3

A repeatable methodology reference for installing Qualys Cloud Agent on SSM-managed EC2 instances with no SSH, no public IP, and no inbound ports — covering the full chain from S3 file staging through agent activation, plus the commands involved and what each one does.

1. Why this deployment method, in plain terms

The goal is to get Cloud Agent software running on each EC2 instance without SSH and without any inbound port open. AWS Systems Manager (SSM) provides the connection; Qualys's installer file still has to get onto the instance somehow, and Qualys's own download page requires an authenticated browser session, so a direct download link isn't available to a command-line tool running on the instance. The workaround is to stage the installer in a private S3 bucket and generate a temporary, signed download link the instance can use once.

2. The full chain, step by step

Step	Tool / Location	What happens
1	Browser (Windows laptop)	Download the Cloud Agent installer (.deb for Ubuntu, .rpm for Amazon Linux) from the Qualys Cloud Agent download page
2	AWS S3 console	Upload the installer file into a private bucket (account regional namespace, all public access blocked)
3	AWS S3 console	Generate a presigned URL — a temporary, signed link granting time-limited access to that one file only
4	AWS EC2 console → Session Manager	Open a browser-based terminal directly on the target instance, authenticated via the instance's IAM role (LindfoSec-SSM-Role) — no SSH, no key
5	wget (in the SSM terminal)	Download the installer file from the presigned URL onto the instance
6	dpkg / rpm (in the SSM terminal)	Install the downloaded package onto the instance's operating system
7	qualys-cloud-agent.sh (in the SSM terminal)	Activate the installed agent against the LindfoSec Qualys subscription using Customer ID, Activation ID, and Web Service URI
8	systemctl status	Confirm the agent service is running locally on the instance
9	Qualys → Cloud Agent → Agent Management	Confirm the instance has checked in and appears as a managed asset

3. The commands, explained

wget — download a file from a URL

A Linux command-line tool that downloads a file directly from a web address, no browser required. Used here to pull the installer from the S3 presigned URL onto the instance.

```
sudo wget -O QualysCloudAgent.deb '<presigned-url>'
```

-O QualysCloudAgent.deb sets the local filename to save as, regardless of what the file was named in S3. sudo is required because writing into the working directory and reaching the network as root-level install prep needs elevated permission on these instances.

dpkg — install a .deb package (Ubuntu/Debian)

Ubuntu and Debian-based systems package software as .deb files. dpkg is the tool that actually installs the contents of that package onto the system — the Linux equivalent of running a downloaded .exe installer on Windows.

```
sudo dpkg --install QualysCloudAgent.deb
```

Amazon Linux uses the .rpm package format instead, installed with a different tool: `sudo rpm -i QualysCloudAgent.rpm`. This applies to LindfoSec-Finance.

qualys-cloud-agent.sh — activate the installed agent

Installing the package puts the software on the machine, but it isn't yet linked to any Qualys account. This script performs that link, telling the agent which subscription (Customer ID), which activation key (Activation ID), and which Qualys platform endpoint (Web Service URI / ServerUri) to report to.

```
sudo /usr/local/qualys/cloud-agent/bin/qualys-cloud-agent.sh \  
  ActivationId=<activation-id> \  
  CustomerId=<customer-id> \  
  ServerUri=https://qagpublic.qg2.apps.qualys.eu/CloudAgent/
```

systemctl status — confirm the agent is running

Checks whether the Cloud Agent background service is actually active on the instance — the local-side confirmation, before checking Qualys itself for the asset to appear.

```
sudo systemctl status qualys-cloud-agent
```

Expected output includes the line: active (running).

4. Troubleshooting encountered along the way

Symptom	Cause	Fix
Permission denied on wget	Command run without sudo	Prefix the command with sudo
400 Bad Request from S3	Presigned URL corrupted or truncated during paste, or shell misinterpreted special characters in the URL	Regenerate the presigned URL; wrap it in single quotes rather than double quotes; build the full command in Notepad first, then paste as one block
403 Forbidden from S3	Presigned URL expired (time-limited by design)	Generate a fresh presigned URL for the same file — no need to re-upload or create a new bucket
Difficult to edit a long pasted command in the SSM terminal	Browser-based SSM terminal doesn't reliably support standard arrow-key/Ctrl cursor shortcuts	Compose the full command in a local text editor (Notepad) first, verify it, then paste the completed line into the terminal in one shot rather than editing in place

5. Deployment status

- LindfoSec-IT-Mgmt — installed, activated, confirmed in Qualys
- LindfoSec-RnD-Workstation (on-prem Windows) — installed, activated, confirmed in Qualys
- LindfoSec-HR — installed; activation in progress
- LindfoSec-Sales-Marketing — pending (same .deb, same steps)
- LindfoSec-Finance — pending (.rpm package, Amazon Linux — different install command)